

FGV EAESP — SÃO PAULO SCHOOL OF BUSINESS ADMINISTRATION

MSC / PHD IN BUSINESS ADMINISTRATION

Time Series Econometrics for Business Research

Lecture 6

Volatility Modelling: ARCH, GARCH and Multivariate Extensions

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Stylised Facts · ARCH · GARCH · EGARCH · GJR-GARCH · Estimation · Diagnostic
Tests · DCC · Value-at-Risk · Expected Shortfall

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1. Stylised Facts of Financial Returns

Let $r_t = \ln(P_t/P_{t-1})$ be the log-return of an asset price. The following properties are observed universally across assets, markets, and frequencies.

- (i) **Near-zero autocorrelation** of r_t (market efficiency).
- (ii) **Volatility clustering**: large changes tend to be followed by large changes; small by small.
- (iii) **Fat tails** (leptokurtosis): extreme returns occur far more often than under normality.
- (iv) **Leverage effect**: negative returns increase volatility more than equal-sized positive returns.
- (v) **Long memory in volatility**: $|r_t|$ and r_t^2 have slowly decaying ACF.

Engle (1982) LM test for ARCH effects. Regress $\hat{\varepsilon}_t^2$ on q lags. Test $H_0 : \alpha_1 = \dots = \alpha_q = 0$ using $TR^2 \sim \chi^2(q)$.

2. ARCH Model

Definition 2.1 (ARCH(q), Engle 1982).

$$r_t = \mu_t + \varepsilon_t, \quad \varepsilon_t = \sigma_t z_t, \quad z_t \stackrel{iid}{\sim} (0, 1),$$

$$\sigma_t^2 = \omega + \alpha_1 \varepsilon_{t-1}^2 + \dots + \alpha_q \varepsilon_{t-q}^2.$$

Constraints: $\omega > 0$, $\alpha_i \geq 0$, $\sum_{i=1}^q \alpha_i < 1$ (covariance stationarity).

The variance σ_t^2 is conditionally heteroskedastic: large past squared errors raise the current conditional variance, producing volatility clustering. The number of lags q required can be large; this motivates the GARCH parsimonisation.

3. GARCH Model

Definition 3.1 (GARCH(p, q), Bollerslev 1986).

$$\sigma_t^2 = \omega + \sum_{i=1}^q \alpha_i \varepsilon_{t-i}^2 + \sum_{j=1}^p \beta_j \sigma_{t-j}^2.$$

Constraints: $\omega > 0$, $\alpha_i, \beta_j \geq 0$, $\sum \alpha_i + \sum \beta_j < 1$.

GARCH(p, q) is an ARMA-type model for ε_t^2 : the squared errors are an ARMA($\max(p, q), p$) process. GARCH(1,1) — the workhorse — typically satisfies $\omega \approx \text{small}$, $\alpha \approx 0.05\text{--}0.15$, $\beta \approx 0.80\text{--}0.90$. The persistence $\alpha + \beta$ is close to 1 for most financial series.

Forecasting GARCH(1,1) variance:

$$\sigma_{t+h|t}^2 = \bar{\sigma}^2 + (\alpha + \beta)^{h-1}(\sigma_{t+1|t}^2 - \bar{\sigma}^2), \quad \bar{\sigma}^2 = \frac{\omega}{1 - \alpha - \beta}.$$

Conditional variance forecasts mean-revert to the unconditional variance $\bar{\sigma}^2$ at rate $(\alpha + \beta)$.

4. Asymmetric GARCH Models**4.1. GJR-GARCH (Glosten, Jagannathan & Runkle, 1993)**

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \gamma \varepsilon_{t-1}^2 \mathbf{1}_{(\varepsilon_{t-1} < 0)} + \beta \sigma_{t-1}^2.$$

The asymmetry parameter $\gamma > 0$ captures the leverage effect: bad news ($\varepsilon_{t-1} < 0$) increases volatility by an additional $\gamma \varepsilon_{t-1}^2$ relative to equal-sized good news.

4.2. EGARCH (Nelson, 1991)

$$\ln \sigma_t^2 = \omega + \beta \ln \sigma_{t-1}^2 + \alpha \left| \frac{\varepsilon_{t-1}}{\sigma_{t-1}} \right| + \gamma \frac{\varepsilon_{t-1}}{\sigma_{t-1}}.$$

Advantages: (i) the log formulation guarantees $\sigma_t^2 > 0$ without sign constraints on parameters; (ii) $\gamma < 0$ captures the leverage effect; (iii) allows testing $H_0 : \gamma = 0$ directly.

5. Estimation and Diagnostic Testing**5.1. Maximum likelihood estimation**

Under the conditional normal distribution:

$$\ell(\theta) = -\frac{T}{2} \ln(2\pi) - \frac{1}{2} \sum_{t=1}^T \left[\ln \sigma_t^2 + \frac{\varepsilon_t^2}{\sigma_t^2} \right].$$

Maximise using BFGS or other gradient methods. Under t -distributed z_t , replace the normal log-likelihood accordingly and estimate the degrees of freedom ν jointly.

5.2. Residual diagnostics

Let $\hat{z}_t = \hat{\varepsilon}_t / \hat{\sigma}_t$ be the standardised residuals. Check:

- Ljung–Box on $\hat{z}_t$: no serial correlation in the mean.
- Ljung–Box on $\hat{z}_t^2$: no remaining ARCH effects.
- Excess kurtosis of $\hat{z}_t$: if significantly positive, use t or GED distribution instead of normal.

6. Multivariate GARCH: Dynamic Conditional Correlation

Definition 6.1 (DCC model, Engle 2002). *Step 1.* Estimate a univariate GARCH for each series i . Obtain standardised residuals $z_{i,t} = \varepsilon_{i,t}/\sigma_{i,t}$.

Step 2. Model dynamic correlations:

$$\mathbf{Q}_t = (1 - a - b)\bar{\mathbf{Q}} + a\mathbf{z}_{t-1}\mathbf{z}_{t-1}' + b\mathbf{Q}_{t-1},$$

$$\mathbf{R}_t = \text{diag}(\mathbf{Q}_t)^{-1/2} \mathbf{Q}_t \text{diag}(\mathbf{Q}_t)^{-1/2}, \quad \boldsymbol{\Sigma}_t = \mathbf{D}_t \mathbf{R}_t \mathbf{D}_t,$$

where $\bar{\mathbf{Q}}$ is the unconditional correlation matrix and $a + b < 1$.

DCC is computationally feasible for large n (two-step estimation), produces positive-definite $\boldsymbol{\Sigma}_t$ by construction, and requires only 2 additional parameters (a, b) for correlation dynamics. Correlations typically spike during financial crises — a key feature for risk management and portfolio construction.

7. Value-at-Risk and Expected Shortfall

Definition 7.1. VaR_α is the loss not exceeded with probability $1 - \alpha$:

$$P(r_{t+1} < -\text{VaR}_\alpha) = \alpha.$$

For GARCH(1,1) with Normal innovations: $\text{VaR}_\alpha = \hat{\mu}_{t+1} + \hat{\sigma}_{t+1}\Phi^{-1}(\alpha)$.

Definition 7.2. $\text{ES}_\alpha = E[-r_{t+1} \mid r_{t+1} < -\text{VaR}_\alpha]$ (Expected Shortfall, also CVaR) is the expected loss conditional on exceeding VaR. ES is a coherent risk measure; VaR is not. Basel III/IV requires ES at 97.5%.

Backtesting. Kupiec (1995): test whether violation frequency equals α . Christoffersen (1998): test independence of violations. GARCH-based VaR substantially outperforms constant-variance VaR in both tests.

8. Core References

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Lecture 6 notes, FGV EAESP. These notes draw on `Modelos_para_a_Variancia` and `Modelos_para_Vol_Mult` by Prof. Pedro Valls Pereira.